Master Thesis

Feature-Aware Numerical Representation for Retrieval-Augmented Medical Tabular Classification (FiTabr)

by

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Abstract

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With the increasing application of artificial intelligence in healthcare, medical tabular data have become an important source for clinical prediction tasks. Such datasets typically contain heterogeneous patient-related variables, including demographic characteristics, laboratory measurements, and physiological indicators. However, learning from medical tabular data remains challenging because of limited sample sizes, heterogeneous feature types, class imbalance, and variation in the predictive relevance of individual variables. Recently, retrieval-augmented tabular learning methods such as TabR have demonstrated promising performance by incorporating information from similar training samples into the prediction process. However, the numerical representation used before retrieval does not explicitly model sample-specific variation in feature emphasis. Therefore, this thesis investigates feature-aware numerical representation for retrieval-augmented medical tabular classification. A feature-aware extension of TabR, termed FiTabR, is proposed by introducing a lightweight attention module before the TabR encoder. The proposed module generates sample-specific attention weights for numerical features and applies residual feature modulation while preserving the original retrieval, contextual aggregation, and prediction mechanisms of TabR. In addition, ablation studies are conducted to examine the contribution of residual preservation, attention scaling, and learned feature weighting to the behavior of the pro-

posed architecture. The learned attention distributions are further analyzed using their mean and standard deviation to characterize feature emphasis and its correspondence with established medical domain knowledge. Experimental results across multiple medical tabular datasets show that FiTabR improves TabR under certain dataset conditions, while the magnitude of improvement varies across datasets. The ablation and attention analyses further indicate that the contribution of the proposed architectural components and the learned feature emphasis is dataset dependent. These findings demonstrate the potential of feature-aware numerical representation for retrieval-augmented tabular learning while also identifying its limitations across different medical data characteristics.

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1. Introduction

clinical information is progressively collected and stored in structured digital form. Many clinical prediction problems are naturally represented as tabular data composed of demographic characteristics, physiological measurements, laboratory values, medical history, and other patient-related variables. Such data support a wide range of predictive tasks, including disease identification, risk estimation, and clinical outcome prediction [1]. However, medical tabular data also present several challenges for predictive modeling. Compared with the large-scale datasets commonly available in computer vision and natural language processing, medical datasets are often relatively small, may contain heterogeneous feature types, and can exhibit substantial differences in class distribution and predictive structure across clinical tasks.

Traditional machine learning methods, particularly tree-based models, remain strong approaches for tabular prediction. Gradient-boosted decision trees such as XGBoost, LightGBM, and CatBoost have demonstrated competitive performance across a wide range of structured-data problems [2, 3, 4]. In contrast, the advantages of deep learning on conventional tabular datasets have been less consistent than in image, text, and speech applications. Previous studies have shown that carefully tuned tree-based models frequently remain competitive with or outperform deep neural networks on typical tabular benchmarks [5]. Nevertheless, recent advances in tabular deep learning have introduced architectures designed specifically to improve the representation of numerical and categorical variables and to model interactions between features and observations [6, 7, 8].

One important direction within this development is retrieval-augmented tab-

ular learning. Rather than generating a prediction solely from parameters learned during training, retrieval-augmented models additionally use information from relevant observations stored in the training set. For a given target sample, similar observations are retrieved and incorporated into the prediction process. This idea connects representation learning with local information from the available data and has motivated several retrieval-based approaches for tabular prediction [9, 10, 11].

TabR is a recent retrieval-augmented deep learning model for tabular data that combines a feed-forward architecture with a nearest-neighbor-like retrieval mechanism [11]. The model first constructs a learned representation of the target observation and candidate training observations. Candidate samples are then retrieved according to similarity in the learned representation space, and information from the retrieved observations is incorporated through a contextual aggregation mechanism before the final prediction is produced. TabR demonstrated that an appropriately designed retrieval mechanism can substantially improve the competitiveness of deep learning for tabular prediction. At the same time, the original TabR study also showed that retrieval is not uniformly beneficial for every dataset, highlighting the importance of dataset characteristics in determining the effectiveness of retrieval-based learning.

The behavior of a retrieval-augmented model depends not only on how candidate observations are retrieved, but also on the representation from which similarity is determined. In TabR, numerical and categorical inputs are encoded into learned sample representations before candidate retrieval is performed. The retrieval component therefore operates on representations whose geometry depends on how the original input variables are transformed. However, the original TabR

architecture does not explicitly introduce sample-specific reweighting of individual numerical variables before the encoder. Its specialized attention-like mechanism is instead primarily associated with the retrieval and aggregation of candidate observations.

This distinction motivates the central question investigated in this thesis. Medical tabular datasets commonly contain numerical measurements whose predictive relevance may vary across prediction tasks and individual samples. An explicit mechanism that allows the numerical representation to adapt to the characteristics of each observation may therefore influence the representation subsequently used by the retrieval model. At the same time, modern neural networks are already capable of learning nonlinear transformations of their inputs, and it cannot be assumed in advance that an additional feature-aware mechanism will consistently improve predictive performance. The effect of such a modification must therefore be evaluated empirically and separated from changes to the retrieval architecture itself.

To investigate this problem, this thesis proposes *FiTabR*, a feature-aware extension of TabR for medical tabular classification. FiTabR introduces a lightweight attention network that operates on numerical features before the TabR encoder. For each input sample, the attention module produces a distribution over the numerical variables. These values are then used to modulate the numerical input through residual feature reweighting. The resulting representation is subsequently processed by the original TabR encoder and retrieval pipeline.

For a numerical feature vector $\mathbf{x}_i$, the FiTabR representation is defined as

$$\tilde{\mathbf{x}}_i = \mathbf{x}_i \odot (1 + \alpha \mathbf{a}_i), \tag{1.1}$$

where $\mathbf{a}_i$ denotes the sample-specific attention distribution, α is the attention scaling coefficient, and $\odot$ denotes element-wise multiplication. The residual term preserves the original numerical input while allowing the learned attention mechanism to modify its magnitude before encoding. Importantly, FiTabR does not replace the retrieval mechanism of TabR. Candidate retrieval, contextual aggregation, and the final prediction mechanism are retained. The architectural change is therefore localized to the numerical representation supplied to the TabR encoder.

This controlled design makes it possible to examine whether changes in pre-retrieval numerical representation can affect the behavior of a retrieval-augmented model without simultaneously redesigning its retrieval mechanism. Accordingly, the primary research question of this thesis is:

> *Can sample-specific numerical feature modulation provide useful improvements to retrieval-augmented tabular classification while preserving the underlying TabR retrieval framework?*

Two additional questions follow from this objective. First, if differences in performance are observed, which components of the proposed representation mechanism are responsible for them? Second, does the learned feature-attention mechanism provide meaningful information about how the model emphasizes numerical clinical variables across different observations?

To address these questions, FiTabR is evaluated on three medical binary classification tasks: Heart Disease, Pima Diabetes, and Stroke Prediction. These datasets provide different experimental conditions in terms of sample size, feature composition, and class distribution. FiTabR is compared directly with the original TabR architecture and with conventional machine learning and deep tabular learn-

ing baselines. The purpose of the cross-dataset evaluation is not to assume that feature-aware representation will improve every medical prediction task, but to determine whether its effect is consistent and to identify the conditions under which the proposed mechanism is beneficial.

The experimental findings provide partial rather than universal support for the initial hypothesis. FiTabR shows its clearest improvement relative to TabR on the Heart Disease dataset, whereas the differences observed on Pima Diabetes are more modest. On the severely imbalanced Stroke Prediction dataset, the proposed representation mechanism does not improve threshold-based minority-class prediction. These results indicate that explicit feature-aware numerical representation can influence retrieval-augmented tabular learning, but that its effectiveness depends on the characteristics of the underlying dataset. In particular, modifying feature representation alone is not sufficient to address learning problems dominated by severe class imbalance.

Predictive comparison between TabR and FiTabR alone is insufficient to determine why the proposed method behaves differently. For this reason, the thesis additionally conducts a component-level ablation study. Three architectural variants are examined: a model without residual feature preservation, a model without the explicit attention-scaling factor, and a model in which learned feature attention is replaced by uniform weighting. These experiments distinguish the contribution of residual preservation, numerical modulation, and sample-specific nonuniform weighting. The ablation results show that the components do not contribute equally across datasets. Residual feature preservation receives the clearest empirical support, particularly on Heart Disease, whereas fixed attention scaling and learned nonuniform feature weighting exhibit more dataset-dependent effects.

The feature-attention mechanism also provides an explicit internal signal that can be examined independently of overall predictive performance. Attention values are therefore analyzed across samples using their mean and standard deviation. The mean attention assigned to a feature describes its average model-assigned emphasis across the dataset, while the standard deviation characterizes how strongly this emphasis varies between individual samples. These patterns are compared with established medical domain knowledge to examine whether the representation learned by FiTabR exhibits clinically plausible correspondence. Attention-based tabular models have previously been used to expose feature-selection and interaction behavior [12]; however, attention values should not automatically be interpreted as causal explanations. Accordingly, the analysis in this thesis treats FiTabR attention as evidence of internal model behavior rather than as a direct estimate of clinical risk or causal feature importance.

Based on this design and experimental analysis, the principal contributions of this thesis are summarized as follows:

1. **Feature-aware extension of retrieval-augmented tabular learning.** FiTabR introduces sample-specific residual modulation of numerical features before the TabR encoder while retaining the original candidate retrieval, contextual aggregation, and prediction mechanisms. This provides a controlled framework for studying feature-aware representation without redesigning the retrieval architecture.

2. **Cross-dataset empirical evaluation of the proposed representation mechanism.** FiTabR is evaluated across three medical tabular classification tasks and compared with TabR, conventional machine learning models, and

deep tabular baselines. The results identify both conditions in which feature-aware representation provides measurable benefit and conditions in which its contribution is limited.

3. **Component-level analysis through architectural ablation.** Residual preservation, attention scaling, and learned nonuniform weighting are evaluated separately to determine how each component contributes to the observed behavior. The analysis shows that the benefit of FiTabR cannot be attributed uniformly to all components and that residual preservation receives the strongest empirical support in the clearest positive case.

4. **Analysis of sample-specific feature emphasis.** The learned attention distributions are examined through their mean and standard deviation and compared with established medical knowledge. This analysis characterizes both population-level feature emphasis and sample-dependent variation while maintaining a distinction between model-assigned attention and clinical importance.

These contributions position FiTabR as a controlled investigation of feature-aware representation within retrieval-augmented tabular learning rather than as a universally superior replacement for TabR. The results are used to identify not only where the proposed representation mechanism provides benefits, but also where its assumptions and capabilities are insufficient. Such analysis is particularly important for medical tabular learning, where dataset size, class distribution, and feature structure can vary substantially between prediction problems.

The remainder of this thesis is organized as follows. Chapter II reviews medical tabular learning, deep learning methods for structured data, retrieval-augmented

tabular learning, the TabR architecture, and feature attention approaches relevant to this work. Chapter III presents the proposed FiTabR architecture, including the problem formulation, feature-attention module, residual feature modulation, and integration with the TabR retrieval framework. Chapter IV describes the datasets, baseline models, experimental protocol, main comparative results, ablation experiments, attention analysis, and discussion of the observed behavior. Finally, Chapter V summarizes the main findings, discusses the limitations of the study, and outlines directions for future research.

2. Research Background

2.1 Classical Machine Learning for Medical Tabular Data

Classical machine learning methods remain strong baselines for structured tabular prediction. In particular, gradient-boosted decision tree (GBDT) methods are widely used because they can model nonlinear relationships and high-order feature interactions while remaining effective on small- and medium-sized tabular datasets. XGBoost introduced a scalable gradient-boosting framework with regularization and efficient tree construction [2], while LightGBM improved computational efficiency through histogram-based learning and leaf-wise tree growth [3]. CatBoost further addressed the treatment of categorical variables through ordered boosting and specialized categorical-feature processing [4].

The continued strength of these approaches is particularly relevant to deep learning research on tabular data. Unlike image and language domains, where deep neural architectures have become dominant, tree-based methods frequently remain highly competitive on conventional tabular benchmarks [5]. This observation makes GBDT models important reference baselines when evaluating new neural architectures for medical tabular prediction.

Local learning provides another relevant perspective. Instead of learning only a global mapping from features to outcomes, local methods make predictions using observations in the neighborhood of a target sample. Early work on local learning demonstrated the general principle of adapting prediction to nearby training observations [13]. Similarly, standard statistical learning methods such as k-nearest

neighbors provide a simple example of instance-based prediction in which similarity between observations directly affects the predicted output [14]. This local-learning perspective is conceptually related to modern retrieval-augmented models, which combine learned representations with information obtained from selected training samples.

2.2 Deep Learning for Tabular Prediction

Recent research has investigated whether neural networks can achieve stronger performance on structured data through architectures specifically designed for tabular inputs. Gorishniy et al. revisited commonly used deep learning models for tabular prediction and demonstrated that carefully configured multilayer perceptrons and Transformer-based architectures can serve as competitive neural baselines [6]. Subsequent work has also emphasized the importance of numerical feature representation, showing that the manner in which continuous variables are embedded can substantially affect the performance of tabular neural networks [7].

Attention mechanisms have been applied to tabular learning in several forms. TabTransformer uses contextual representations of categorical variables to model relationships among features [15]. TabNet introduces sequential attentive feature selection, allowing the model to focus on subsets of variables at different decision steps [12]. SAINT extends attention beyond feature interactions by applying attention across both rows and columns and combining this mechanism with contrastive pre-training [8]. These methods demonstrate that attention can be used to adapt representations to the structure of tabular data rather than treating the input only through a standard feed-forward transformation.

Despite these developments, deep learning does not consistently dominate tree-

based methods on typical tabular datasets. Grinsztajn et al. showed that tree-based models retain important advantages on many medium-sized structured datasets, particularly under conditions commonly encountered in practical tabular learning [5]. Consequently, recent research has explored mechanisms beyond conventional feed-forward representation learning, including improved numerical embeddings, attention-based feature interaction, and retrieval of information from other observations.

2.3 Retrieval-Augmented Tabular Learning

Retrieval-augmented learning extends conventional predictive models by allowing information from other training observations to participate directly in the prediction of a target sample. The underlying idea is related to local and instance-based learning, but modern retrieval-augmented methods combine this principle with learned neural representations. Instead of relying exclusively on predefined distances in the original input space, these methods learn a representation in which relevant observations can be identified and their information subsequently incorporated into prediction.

Several approaches have investigated interactions among observations for tabular prediction. The Retrieval and Interaction Machine (RIM) combines retrieval with interaction mechanisms to exploit information from related training instances [9]. Other work has investigated neighborhood-based representation propagation as a means of enriching the representation of tabular observations using information from nearby samples [16]. Self-attention between datapoints has also been proposed as an alternative to architectures that process each observation independently, demonstrating that relationships among samples can provide additional

predictive information [10]. Related memory-based approaches, including Hopular, further demonstrate the potential of associating tabular observations with information stored in the training data [17, 18].

TabR provides a more direct and computationally efficient realization of retrieval-augmented tabular deep learning. The model combines a feed-forward network with a nearest-neighbor-like retrieval component operating in a learned representation space [11]. For each target observation, TabR constructs a query representation, identifies relevant candidate observations from the training set, and incorporates their representations and labels through an attention-like contextual aggregation mechanism. The resulting contextual information is combined with the target representation before final prediction.

The TabR study demonstrated that retrieval can substantially strengthen tabular deep learning on a range of benchmarks while using a comparatively simple architecture [11]. Importantly, however, the effectiveness of retrieval was shown to depend on the dataset. The study also identified numerical representation and retrieval as distinct but interacting components of tabular deep learning, suggesting that the representation on which similarity is computed is itself an important architectural consideration.

FiTabR builds directly on this observation. Rather than modifying candidate selection, contextual aggregation, or the prediction mechanism, FiTabR changes the numerical representation supplied to the TabR encoder. A lightweight attention network generates a sample-specific distribution over numerical variables, and these values are used for residual modulation before encoding. The subsequent TabR retrieval pipeline is retained. FiTabR therefore provides a controlled setting for studying whether explicit sample-specific numerical modulation before repre-

sentation learning can affect the behavior of a retrieval-augmented tabular model.

2.4 Attention and Explainability in Medical Machine Learning

Interpretability is an important consideration in medical machine learning because predictive performance alone may be insufficient for understanding how a model arrives at its outputs. General-purpose post-hoc explanation methods have therefore been widely applied to complex predictive models. LIME explains individual predictions by constructing locally interpretable approximations of a model around a particular observation [19], whereas SHAP uses concepts from cooperative game theory to assign feature-attribution values to model predictions [20]. More broadly, the application of machine learning in medicine has emphasized the importance of considering model interpretation, validation, and clinical context alongside predictive performance [1].

Attention-based tabular models provide another source of information about internal model behavior. TabNet, for example, exposes feature-selection masks that indicate which variables are emphasized during different decision steps [12]. In similarity-based learning, interpretable prediction has also been studied through mechanisms that relate predictions to nearby training observations. Wang and Sabuncu showed that a flexible Nadaraya–Watson prediction head can provide similarity-based explanations while supporting calibrated classification [21].

FiTabR combines these two perspectives at the architectural level. TabR already provides an instance-retrieval mechanism through which selected training observations contribute contextual information to prediction. FiTabR additionally

produces sample-specific attention values over the numerical features before encoding. These values create an explicit feature-level signal that can be examined across observations.

However, attention values should not automatically be interpreted as causal feature importance or clinical risk estimates. In this thesis, they are treated as indicators of the internal allocation of emphasis within the FiTabR representation mechanism. Mean attention values are used to characterize the average emphasis assigned to individual numerical variables, while their standard deviations are used to examine how that emphasis varies across samples. The resulting patterns are subsequently compared with established medical knowledge as a behavioral analysis of the model rather than as a causal explanation.

2.5 Research Gap and Positioning of FiTabR

Existing work has therefore addressed several complementary aspects of tabular learning. Gradient-boosted decision trees provide strong non-neural baselines [2, 3, 4]; attention-based neural architectures improve feature representation and interaction modeling [15, 12, 8]; and retrieval-augmented approaches incorporate information from other training observations [9, 10, 11].

Within the methods considered in this thesis, however, the effect of explicitly introducing sample-specific modulation of numerical variables before the TabR encoder and retrieval representation remains insufficiently characterized. TabR learns the representation on which candidate similarity is evaluated, but does not explicitly introduce a separate sample-specific numerical feature-modulation stage before encoding. Conversely, attention-based tabular models provide mechanisms for feature interaction or selection, but are not designed as controlled modifications of

the TabR retrieval framework.

FiTabR addresses this gap by introducing feature-aware residual modulation of numerical inputs before the TabR encoder while preserving the original retrieval, contextual aggregation, and prediction mechanisms. This design isolates the representation-level modification from changes to the retrieval architecture and enables three related questions to be studied: whether the modification improves predictive behavior, which components of the feature-modulation mechanism contribute to any observed differences, and how the learned feature emphasis behaves across medical datasets.

The contribution is therefore not based on the assumption that feature-aware modulation is universally superior to the original TabR representation. Instead, FiTabR is evaluated as a controlled architectural extension whose effectiveness may depend on the characteristics of the dataset. Ablation experiments and attention analysis are subsequently used to determine the conditions under which the proposed representation mechanism provides measurable value and to identify its limitations.

3. Methodology

3.1 Problem Formulation

This study proposes FiTabR as an extension of the TabR retrieval-augmented tabular learning model for binary classification on medical tabular datasets. The methodology is designed to isolate the effect of feature-aware numerical representation learning while preserving the original TabR retrieval mechanism. Therefore, the proposed model modifies only the input representation stage before the TabR encoder, allowing a controlled comparison between TabR and FiTabR under the same retrieval, aggregation, and prediction framework.

Proposed method considers supervised binary classification on medical tabular datasets. Let the dataset be denoted as $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^{N}$, where x_i is the feature vector of the i-th sample, $y_i \in \{0, 1\}$ is the corresponding binary class label, and N is the total number of samples. The binary label represents the target clinical outcome, such as the presence or absence of a disease.

Each input sample consists of numerical and categorical components, represented as $x_i = (x_i^{\text{num}}, x_i^{\text{cat}})$. Here, x_i^{num} denotes the numerical feature vector and x_i^{cat} denotes the categorical feature component. The objective is to learn a prediction function $f(x_i) \to \hat{y}_i$, where $\hat{y}_i$ represents the predicted class probability or class label.

Medical tabular datasets often contain limited samples, heterogeneous feature types, and variables with different levels of clinical relevance. Therefore, the model should not only produce accurate predictions but also construct meaningful representations for retrieval-based reasoning.

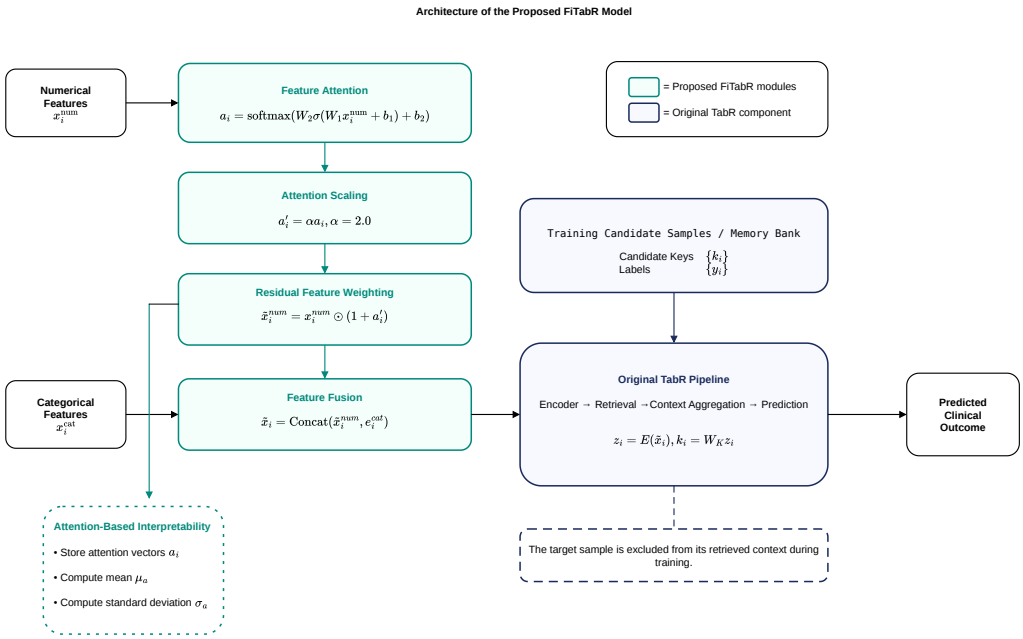


Figure 3.1: Architecture of the proposed FiTabR model. FiTabR applies feature attention, attention scaling, and residual feature weighting to numerical features before feature fusion and the original TabR retrieval and prediction pipeline.

3.2 Base Retrieval Model

This study uses TabR as the base retrieval-augmented tabular learning architecture. TabR combines neural representation learning with a retrieval mechanism that uses similar training samples as contextual information during prediction. In the proposed FiTabR model, the original TabR retrieval architecture is preserved. The main modification is introduced before the TabR encoder, where numerical features are adaptively reweighted using a feature-attention module. Therefore, this section focuses on the proposed feature-aware representation mechanism rather than re-describing the full TabR architecture.

3.3 Proposed FiTabR Architecture

FiTabR extends TabR by introducing a feature-attention module before the TabR encoder. The motivation for this modification is that not all numerical medical features contribute equally to prediction or patient similarity. In clinical datasets, some variables may be highly informative for a specific disease prediction task, while others may be weakly related or noisy. Therefore, treating all numerical features uniformly before retrieval may reduce the quality of the retrieved neighborhood.

The proposed FiTabR pipeline follows the sequence $x_i^{\text{num}} \rightarrow a_i \rightarrow \tilde{x}_i^{\text{num}} \rightarrow \tilde{x}_i \rightarrow z_i \rightarrow k_i \rightarrow \hat{y}_i$. First, the numerical input is passed through a feature-attention module to obtain the attention vector a_i. The attention vector is then used to generate the weighted numerical representation $\tilde{x}_i^{\text{num}}$. After feature fusion, the final feature-aware representation $\tilde{x}_i$ is passed into the TabR architecture for retrieval-based prediction.

This design keeps the main TabR retrieval structure unchanged while adding a feature-aware preprocessing stage before representation learning and retrieval.The proposed modification is motivated by the assumption that retrieval quality in TabR depends strongly on the representation used to compute candidate similarity. In medical tabular datasets, numerical variables often differ in diagnostic relevance, measurement scale, and noise level. If all numerical features are encoded uniformly, weak or noisy features may affect the learned representation and consequently degrade neighborhood retrieval. Therefore, FiTabR introduces a feature-attention module before the TabR encoder to adaptively emphasize informative numerical variables while preserving the original retrieval mechanism.

3.4 Feature-Attention Module

The feature-attention module is applied to the numerical feature vector x_i^{num}. The attention vector a_i is computed as

$$a_i = \text{softmax}\left(W_2\sigma\left(W_1 x_i^{\text{num}} + b_1\right) + b_2\right), \tag{3.1}$$

where W_1 and W_2 are trainable weight matrices, b_1 and b_2 are bias terms, and $\sigma(\cdot)$ denotes the ReLU activation function. The softmax function normalizes the attention scores across numerical features, allowing the model to assign sample-specific importance to each numerical variable.

The trainable matrices W_1 and W_2, together with the bias terms b_1 and b_2, allow the model to learn nonlinear relationships between numerical features and their importance. Therefore, the feature-attention module can adaptively emphasize features that are more relevant for constructing the retrieval representation.

3.5 Attention Scaling and Residual Feature Weighting

After computing the attention vector a_i, FiTabR scales it using $a_i' = \alpha a_i$, where α is the attention scaling coefficient. In the current implementation, $\alpha = 2.0$. This scaling step increases the influence of the attention vector before it is applied to the original numerical input. Direct attention weighting may suppress original clinical measurements too aggressively, especially in small medical datasets. The residual form preserves the original feature values and only amplifies features selected by the learned attention vector. The value =2.0 is used as an implementation-level scaling coefficient and is examined empirically in the experimental analysis.

The scaled attention vector a'_i is then applied to the original numerical input through residual feature weighting:

$$\tilde{x}_i^{\text{num}} = x_i^{\text{num}} \odot \left(1 + a'_i\right), \tag{3.2}$$

where $\odot$ denotes element-wise multiplication. The residual term $1 + a'_i$ preserves the original numerical feature values while allowing the model to amplify selected features. Thus, the attention mechanism acts as a learned feature enhancement step rather than replacing the original input.

3.6 Feature Fusion

After residual feature weighting, the weighted numerical representation $\tilde{x}_i^{\text{num}}$ is combined with the encoded categorical representation e_i^{cat}. The fused feature-aware representation is defined as $\tilde{x}_i = \text{Concat}(\tilde{x}_i^{\text{num}}, e_i^{\text{cat}})$. This representation is then passed to the TabR encoder.

The role of this step is to ensure that both numerical and categorical information are available to the TabR encoder. FiTabR modifies the numerical feature representation through attention, while categorical features are preserved through their encoded representation.

3.7 Integration with TabR Retrieval

After feature weighting and feature fusion, the resulting representation $\tilde{x}_i$ is passed into the original TabR architecture. The TabR encoder produces the latent representation $z_i = E(\tilde{x}_i)$, which is projected into the retrieval key space as $k_i = W_K z_i$, where W_K is a learnable key projection matrix.

For a target sample q, the similarity between the target representation and a candidate sample i is computed using the negative squared Euclidean distance:

$$S(z_q, z_i) = - \|k_q - k_i\|_2^2 . \tag{3.3}$$

Candidate samples with higher similarity scores are retrieved as contextual samples. FiTabR does not modify the similarity function, neighbor retrieval mechanism, contextual value construction, or prediction head of TabR. Instead, it changes the representation that enters the retrieval pipeline by applying feature-aware numerical weighting before encoding.

For each retrieved candidate sample, the contextual value representation is computed as

$$V(z_q, z_i, y_i) = W_Y(y_i) + T(k_q - k_i), \tag{3.4}$$

where $W_Y(y_i)$ denotes the label embedding of the retrieved candidate sample and $T(\cdot)$ is a learnable transformation of the difference between the target key and the candidate key. The retrieved contextual values are aggregated into C_q, and the prediction head produces the final output $\hat{y}_q = P(z_q, C_q)$.

This design isolates the contribution of FiTabR to the feature-representation stage. As a result, any performance difference between TabR and FiTabR can be interpreted as the effect of adding feature-aware numerical weighting before retrieval.

3.8 Attention-Based Interpretability

FiTabR stores the attention vectors produced by the feature-attention module. Since each attention vector a_i is computed for an individual sample, these values can be analyzed to understand which numerical features receive higher emphasis before retrieval.

To summarize feature-attention behavior across the dataset, FiTabR computes the mean attention score $\mu_a = \frac{1}{N}\sum_{i=1}^{N} a_i$. It also computes the standard deviation $\sigma_a = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(a_i - \mu_a)^2}$. The mean attention score indicates the average attention assigned to each numerical feature, while the standard deviation shows how much attention varies across samples.

These attention statistics provide an additional interpretability signal. However, they should not be interpreted as complete clinical explanations. Instead, they indicate which numerical features the model emphasizes before retrieval and should be analyzed together with predictive performance, retrieved-neighbor behavior, and medical domain knowledge.

3.9 Methodological Summary

FiTabR preserves the retrieval-based structure of TabR while introducing a feature-attention module before the retrieval stage. The proposed method first computes sample-specific attention scores for numerical features, then applies residual feature weighting to construct a feature-aware numerical representation. The weighted representation is then passed into the TabR retrieval architecture, where similar samples are retrieved and used as contextual information for prediction.

This design allows FiTabR to preserve the original retrieval-augmented learn-

ing behavior of TabR while introducing a lightweight mechanism for feature-aware representation construction. As a result, the model can emphasize informative numerical variables before retrieval and provide attention statistics that support interpretation.

4. Experimental Setup

FiTabR was evaluated on three medical tabular binary-classification datasets. The experimental protocol covered data partitioning, preprocessing, baseline selection, hyperparameter optimisation, repeated training, and final test-set evaluation. All models were evaluated using the same data partitions and metric implementations. For the direct architectural comparison, FiTabR and the reference retrieval-based model additionally used the same candidate memory, retrieval configuration, optimisation objective, and random seeds. The implementation was developed using the experimental framework of the reference architecture [11].

4.1 Datasets

Experiments were conducted on the Heart Disease, Pima Indians Diabetes, and Stroke Prediction datasets [22, 23, 24]. The datasets represent different clinical prediction tasks and vary in sample size, feature composition, and class distribution. All tasks were formulated as supervised binary classification. For an observation x_i, the corresponding target was denoted by $y_i \in \{0, 1\}$.

4.1.1 Heart Disease

The Heart Disease dataset contains 918 observations, six numerical features, and five categorical features. The objective is to predict the presence or absence of heart disease from demographic characteristics, symptoms, and clinical measurements. The processed data were divided into 642 training samples, 138 validation samples, and 138 test samples.

Table 4.1: Summary of the Medical Tabular Datasets

Dataset	Samples	Numerical	Categorical	Task
Heart Disease	918	6	5	Binary
Pima Diabetes	768	8	0	Binary
Stroke Prediction	5,110	5	5	Binary

4.1.2 Pima Indians Diabetes

The Pima Indians Diabetes dataset contains 768 observations and eight numerical features. The prediction target indicates whether a patient has diabetes based on diagnostic measurements, including glucose concentration, blood pressure, body mass index, insulin level, age, and pregnancy history. The processed data were divided into 537 training samples, 115 validation samples, and 116 test samples.

4.1.3 Stroke Prediction

The Stroke Prediction dataset contains 5,110 observations, five numerical features, and five categorical features. The objective is to predict stroke occurrence from demographic information, medical history, lifestyle indicators, and physiological measurements. The processed data were divided into 3,577 training samples, 766 validation samples, and 767 test samples. Because the positive class is substantially less frequent than the negative class, threshold-independent and class-sensitive metrics were included in the evaluation.

The selected datasets reflect common constraints in medical tabular learning, including limited sample size, heterogeneous variables, unequal feature scales, missing observations, and class imbalance. Evaluating the same architecture across the three datasets therefore provides evidence regarding the consistency of the proposed feature-aware representation mechanism under different clinical data conditions.

4.2 Preprocessing

A common preprocessing pipeline was applied to all models. Each preprocessing operation was fitted using only the training partition and was subsequently applied to the validation and test partitions without refitting. This procedure prevented validation or test information from influencing model training.

Each dataset was divided into disjoint training, validation, and test subsets using an approximate ratio of 70:15:15. Stratified partitioning was used to preserve the class distribution across the subsets. The training subset was used to estimate model parameters, the validation subset was used for hyperparameter selection and early stopping, and the test subset was reserved for final evaluation.

Numerical features were processed using quantile transformation. This operation reduced the effect of extreme values and differences in feature scale. The transformation was estimated from the training subset and then applied to the validation and test subsets. Categorical variables were integer encoded using mappings derived from the training data. The same mappings were retained for the remaining subsets.

Missing numerical values were imputed using statistics calculated from the training subset. Missing categorical values were represented by a dedicated category or processed according to the categorical policy supported by the corresponding model. Validation- and test-set statistics were not used during imputation.

Synthetic oversampling was not applied in the principal experiments. The original class distributions were preserved to avoid introducing artificial observations into the retrieval candidate memory. The effects of class imbalance were assessed using recall, F1-score, and receiver operating characteristic area under the

curve (ROC–AUC), in addition to accuracy.

For the retrieval-based models, the candidate memory contained only training observations. Validation and test observations were not inserted into the memory. During training, the current target observation was excluded from its own candidate context. These restrictions prevented self-retrieval and label leakage.

4.3 Baselines

FiTabR was compared with gradient-boosted decision trees, conventional neural networks, attention-based deep tabular models, and the original retrieval-based architecture. The selected baselines cover the principal model families commonly used for supervised tabular classification.

4.3.1 Gradient-Boosted Decision Trees

XGBoost [2], LightGBM [3], and CatBoost [4] were used as tree-based baselines. These methods provide strong non-deep-learning reference points for small and medium-sized tabular datasets. CatBoost was also included because the Heart Disease and Stroke Prediction datasets contain categorical variables.

4.3.2 Deep Tabular Models

A multilayer perceptron (MLP) was used as the basic neural-network baseline.[6] The MLP processes each observation independently and does not use feature attention or retrieved training instances. FT-Transformer [25] represents individual features as tokens and applies self-attention to model interactions among features. SAINT [8] provides an additional attention-based comparison for tabular representation learning.

4.3.3 Retrieval-Based Baseline

The original retrieval-based architecture TabR [11] was used as the principal architectural baseline. FiTabR preserves its candidate retrieval, context aggregation, and prediction framework while modifying the input representation through feature attention, residual feature weighting, and feature fusion. This design supports a controlled assessment of whether the proposed representation module improves similarity learning and downstream classification.

4.4 Evaluation Metrics

Performance was evaluated using accuracy, precision, recall, F1-score, ROC–AUC, and binary cross-entropy. The first four metrics were computed from the classification decisions, whereas ROC–AUC and binary cross-entropy were computed from predicted probabilities.

4.4.1 Accuracy

Accuracy measures the proportion of correctly classified observations:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}, \tag{4.1}$$

where TP, TN, FP, and FN denote true positives, true negatives, false positives, and false negatives, respectively.

4.4.2 Precision and Recall

Precision measures the proportion of positive predictions that are correct:

$$\text{Precision} = \frac{TP}{TP + FP}. \tag{4.2}$$

Recall measures the proportion of positive observations correctly identified:

$$\text{Recall} = \frac{TP}{TP + FN}. \tag{4.3}$$

Recall is particularly relevant to medical classification because false-negative predictions may cause affected patients to be missed.

4.4.3 F1-Score

The F1-score is the harmonic mean of precision and recall:

$$\text{F1} = 2\frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}. \tag{4.4}$$

It provides a balanced measure when both false-positive and false-negative errors are relevant.

4.4.4 ROC–AUC

ROC–AUC measures the ability of a model to rank positive observations above negative observations across classification thresholds. A value of 0.5 indicates random ranking, whereas a value of 1.0 indicates perfect class discrimination. This metric was included because it is less dependent on a single decision threshold than accuracy or F1-score.

4.4.5 Binary Cross-Entropy

Binary cross-entropy evaluates the quality of the predicted probabilities:

$$\mathcal{L}_{\text{BCE}} = -\frac{1}{N}\sum_{i=1}^{N}\left[y_i \log(\hat{p}_i) + (1 - y_i)\log(1 - \hat{p}_i)\right], \tag{4.5}$$

where y_i is the true label and $\hat{p}_i$ is the predicted probability of the positive class. Lower values indicate better probabilistic predictions.

For the imbalanced Stroke Prediction dataset, ROC–AUC, recall, and F1-score were considered together with accuracy. This avoided drawing conclusions from majority-class performance alone.

4.5 Implementation Details

The models were implemented in Python using a shared experimental framework. Dataset processing and metric computation were based on scikit-learn [26]. Hyperparameter optimisation was performed using Optuna with the Tree-structured Parzen Estimator sampler [27]. The repository configurations used 200 optimisation trials, including 20 initial startup trials. Hyperparameter selection was based exclusively on validation performance.

For neural models, the search space included learning rate, weight decay, hidden representation dimension, number of processing blocks, batch size, and dropout. Attention-based models additionally included attention dropout and attention-specific representation parameters. For FiTabR, parameters associated with feature attention, residual scaling, and feature fusion were included in the optimisation procedure.

For the tree-based baselines, the search space included learning rate, tree depth, number of boosting iterations, regularisation coefficients, bagging parameters, and leaf-estimation settings. Early stopping was applied using the validation subset.

Neural models were trained by mini-batch gradient-based optimisation using binary cross-entropy as the objective function. The candidate memory for retrieval-based training consisted of the complete training partition. The retrieval configuration was kept fixed between FiTabR and the architectural baseline so that the

primary experimental difference was the proposed feature-aware input representation.

Each selected configuration was evaluated using 15 random seeds, numbered from 0 to 14. The trained models were divided into three groups:

$$\{0, 1, 2, 3, 4\}, \quad \{5, 6, 7, 8, 9\}, \quad \{10, 11, 12, 13, 14\}. \tag{4.6}$$

Predicted probabilities were averaged within each group to form three five-model ensembles. For a metric m, the mean over R runs was calculated as

$$\bar{m} = \frac{1}{R} \sum_{r=1}^{R} m_r, \tag{4.7}$$

and the standard deviation was calculated as

$$s_m = \sqrt{\frac{1}{R-1} \sum_{r=1}^{R} (m_r - \bar{m})^2}. \tag{4.8}$$

The direct comparison between FiTabR and the architectural baseline used identical preprocessing, data partitions, candidate memory, retrieval settings, optimisation objective, early-stopping procedure, random seeds, and evaluation metrics. The principal difference was the feature-aware representation module introduced before the encoder.

All experiments were conducted using Python 3.9.16 and PyTorch 1.13.1 on a system equipped with an Intel Core i7-7700K CPU operating at 4.20 GHz, 32 GB of RAM, and an NVIDIA GeForce GTX 1050 GPU with 2 GB of dedicated memory. The software environment included CUDA 11.7.99, scikit-learn 1.2.2, XGBoost 1.7.4, LightGBM 3.3.5, CatBoost 1.1.1, and Optuna 3.1.0.

5. Results and Discussion

This section evaluates the predictive performance of FiTabR against the original TabR architecture and representative machine-learning and deep-learning baselines on the Heart Disease, Pima Diabetes, and Stroke Prediction datasets. The baselines include the gradient-boosted tree methods CatBoost, XGBoost, and LightGBM [4, 2, 3], together with an MLP and the tabular deep-learning models SAINT and FT-Transformer [8, 6]. TabR is included as the direct retrieval-augmented architectural baseline [11]. The reported values correspond to ensemble evaluation and are expressed as mean ± standard deviation. Precision, recall, and F1-score in the aggregate comparison are weighted averages unless otherwise stated. Accuracy, precision, recall, F1-score, and ROC-AUC are maximized, whereas lower cross-entropy indicates better probabilistic prediction.

5.1 Overall Comparison With All Baselines

Tables 5.1–5.3 summarize the performance of the evaluated models. The results show that no single model dominates across all datasets and evaluation criteria. The relative performance of tree-based and deep-learning methods varies across datasets, while FiTabR is most competitive on the Heart Disease dataset.

Table 5.1: Performance comparison on the Heart Disease dataset. Results are reported as mean ± standard deviation.

Model	Accuracy	Precision	Recall	F1-score	ROC-AUC	Cross-Entropy
CatBoost	0.8889 ± 0.0042	0.8896 ± 0.0043	0.8889 ± 0.0042	0.8890 ± 0.0042	0.9334 ± 0.0001	0.3390 ± 0.0060
XGBoost	0.8841 ± 0.0000	0.8856 ± 0.0008	0.8841 ± 0.0000	0.8843 ± 0.0001	0.9341 ± 0.0030	0.3481 ± 0.0146
LightGBM	0.8841 ± 0.0000	0.8847 ± 0.0000	0.8841 ± 0.0000	0.8842 ± 0.0000	0.9428 ± 0.0005	0.3281 ± 0.0086
MLP	0.8720 ± 0.0042	0.8729 ± 0.0045	0.8720 ± 0.0042	0.8722 ± 0.0042	0.9351 ± 0.0007	0.3350 ± 0.0031
SAINT	0.8792 ± 0.0221	0.8814 ± 0.0205	0.8792 ± 0.0221	0.8795 ± 0.0220	0.9390 ± 0.0011	0.3322 ± 0.0186
FT-Transformer	0.8696 ± 0.0072	0.8716 ± 0.0064	0.8696 ± 0.0072	0.8698 ± 0.0072	0.9344 ± 0.0037	0.3582 ± 0.0098
TabR	0.8841 ± 0.0000	0.8880 ± 0.0000	0.8841 ± 0.0000	0.8844 ± 0.0000	$\mathbf{0.9510 \pm 0.0006}$	0.3586 ± 0.0267
FiTabR	$\mathbf{0.8937 \pm 0.0042}$	$\mathbf{0.8977 \pm 0.0043}$	$\mathbf{0.8937 \pm 0.0042}$	$\mathbf{0.8940 \pm 0.0042}$	$\mathbf{0.9510 \pm 0.0009}$	$\mathbf{0.3038 \pm 0.0097}$

On Heart Disease, FiTabR achieves the strongest overall results among the evaluated models. In particular, it obtains the highest mean accuracy, precision, recall, and F1-score, while its ROC-AUC is effectively tied with TabR at approximately 0.951. FiTabR also records the lowest cross-entropy, suggesting improved confidence and probabilistic prediction. CatBoost is the closest conventional baseline in terms of accuracy, whereas LightGBM obtains the strongest ROC-AUC among the non-retrieval models. The result therefore indicates that the proposed feature-aware representation is particularly effective on this dataset.

Table 5.2: Performance comparison on the Pima Diabetes dataset. Results are reported as mean ± standard deviation.

Model	**Accuracy**	**Precision**	**Recall**	**F1-score**	**ROC-AUC**	**Cross-Entropy**
CatBoost	0.7586 ± 0.0086	0.7550 ± 0.0083	0.7586 ± 0.0086	0.7554 ± 0.0080	0.7918 ± 0.0035	0.6831 ± 0.0029
XGBoost	0.7672 ± 0.0000	0.7636 ± 0.0002	0.7672 ± 0.0000	0.7610 ± 0.0011	0.8078 ± 0.0034	0.5895 ± 0.0236
LightGBM	0.7500 ± 0.0149	0.7467 ± 0.0159	0.7500 ± 0.0149	0.7437 ± 0.0132	0.8086 ± 0.0060	$\mathbf{0.5253 \pm 0.0023}$
MLP	0.7644 ± 0.0050	0.7633 ± 0.0071	0.7644 ± 0.0050	0.7635 ± 0.0065	0.7918 ± 0.0054	0.5659 ± 0.0185
SAINT	0.7529 ± 0.0100	0.7638 ± 0.0073	0.7529 ± 0.0100	0.7558 ± 0.0095	0.8054 ± 0.0003	0.5339 ± 0.0048
FT-Transformer	$\mathbf{0.7701 \pm 0.0263}$	$\mathbf{0.7774 \pm 0.0203}$	$\mathbf{0.7701 \pm 0.0263}$	$\mathbf{0.7722 \pm 0.0250}$	$\mathbf{0.8135 \pm 0.0071}$	0.5337 ± 0.0223
TabR	0.7529 ± 0.0050	0.7515 ± 0.0062	0.7529 ± 0.0050	0.7520 ± 0.0058	0.7963 ± 0.0072	0.5511 ± 0.0120
FiTabR	0.7586 ± 0.0086	0.7595 ± 0.0099	0.7586 ± 0.0086	0.7589 ± 0.0092	0.7953 ± 0.0047	0.5474 ± 0.0044

The Pima Diabetes dataset produces a more competitive ordering among the model families. FT-Transformer achieves the highest mean accuracy, precision, recall, F1-score, and ROC-AUC, while LightGBM obtains the lowest cross-entropy. FiTabR therefore does not provide the best absolute performance on this dataset. Nevertheless, its comparison with TabR remains important because the two architectures differ primarily in the proposed feature-aware input representation. FiTabR improves the main threshold-dependent classification metrics and reduces cross-entropy, while its ROC-AUC remains close to that of TabR.

On Stroke Prediction, the aggregate accuracy of almost all models is close to 0.95. LightGBM achieves the highest mean accuracy and recall, XGBoost obtains the highest precision and F1-score, and FT-Transformer provides the highest

Table 5.3: Performance comparison on the Stroke Prediction dataset. Results are reported as mean ± standard deviation.

Model	Accuracy	Precision	Recall	F1-score	ROC-AUC	Cross-Entropy
CatBoost	0.9500 ± 0.0008	0.9033 ± 0.0000	0.9500 ± 0.0008	0.9261 ± 0.0004	0.8294 ± 0.0168	0.3105 ± 0.0525
XGBoost	0.9513 ± 0.0008	$\mathbf{0.9407 \pm 0.0125}$	0.9513 ± 0.0008	$\mathbf{0.9300 \pm 0.0016}$	0.8201 ± 0.0024	0.2259 ± 0.0678
LightGBM	$\mathbf{0.9518 \pm 0.0013}$	0.9376 ± 0.0296	$\mathbf{0.9518 \pm 0.0013}$	0.9294 ± 0.0031	0.8199 ± 0.0023	$\mathbf{0.1640 \pm 0.0007}$
MLP	0.9505 ± 0.0000	0.9034 ± 0.0000	0.9505 ± 0.0000	0.9263 ± 0.0000	0.8220 ± 0.0087	0.1738 ± 0.0129
SAINT	0.9505 ± 0.0000	0.9034 ± 0.0000	0.9505 ± 0.0000	0.9263 ± 0.0000	0.7368 ± 0.0528	0.2042 ± 0.0067
FT-Transformer	0.9505 ± 0.0000	0.9034 ± 0.0000	0.9505 ± 0.0000	0.9263 ± 0.0000	$\mathbf{0.8328 \pm 0.0036}$	0.1663 ± 0.0049
TabR	0.9505 ± 0.0000	0.9034 ± 0.0000	0.9505 ± 0.0000	0.9263 ± 0.0000	0.8136 ± 0.0078	0.1780 ± 0.0050
FiTabR	0.9505 ± 0.0000	0.9034 ± 0.0000	0.9505 ± 0.0000	0.9263 ± 0.0000	0.8095 ± 0.0126	0.1750 ± 0.0091

ROC-AUC. FiTabR and TabR obtain identical accuracy, precision, recall, and F1-score, while FiTabR slightly reduces cross-entropy from 0.1780 to 0.1750. In contrast, the mean ROC-AUC decreases from 0.8136 for TabR to 0.8095 for FiTabR. The high aggregate accuracy, however, requires further analysis because the Stroke dataset is strongly imbalanced.

5.1.1 Direct TabR–FiTabR Comparison

Since FiTabR is constructed as an extension of TabR, the direct comparison between these two models provides the most controlled evaluation of the proposed feature-aware representation. The retrieval, aggregation, and prediction framework is retained, while FiTabR introduces feature-dependent representation learning before the TabR encoder.

On Heart Disease, FiTabR improves accuracy from 0.8841 ± 0.0000 to 0.8937 ± 0.0042, corresponding to an absolute increase of approximately 0.96 percentage points. Precision increases from 0.8880 to 0.8977, recall from 0.8841 to 0.8937, and F1-score from 0.8844 to 0.8940. The two models obtain essentially identical mean ROC-AUC (0.9510), whereas FiTabR reduces cross-entropy from 0.3586 ± 0.0267 to 0.3038 ± 0.0097. Thus, the principal improvement on Heart Disease occurs in final classification behavior and probabilistic prediction rather than in ranking performance.

On Pima Diabetes, FiTabR improves accuracy from 0.7529±0.0050 to 0.7586± 0.0086 and F1-score from 0.7520 ± 0.0058 to 0.7589 ± 0.0092. Precision and recall exhibit the same direction of change, and cross-entropy decreases from 0.5511 to 0.5474. The ROC-AUC difference is small and favors TabR, with mean values of 0.7963 and 0.7953, respectively. Therefore, the feature-aware representation provides modest gains in threshold-dependent metrics but does not improve ranking performance on Pima Diabetes.

On Stroke Prediction, the two retrieval models obtain identical mean accuracy, precision, recall, and F1-score. FiTabR produces a small reduction in cross-entropy but a lower mean ROC-AUC. The direct comparison therefore shows that the benefit of feature-aware representation is dataset-dependent, with the clearest improvement occurring on Heart Disease.

5.1.2 Heart Disease Results

The Heart Disease dataset provides the strongest quantitative support for FiTabR. The proposed model achieves an accuracy of 0.8937 ± 0.0042, exceeding TabR, CatBoost, XGBoost, LightGBM, MLP, SAINT, and FT-Transformer. The same ordering is observed for precision, recall, and F1-score. In particular, FiTabR obtains an F1-score of 0.8940 ± 0.0042, compared with 0.8890 ± 0.0042 for CatBoost and 0.8844 ± 0.0000 for TabR.

The ROC-AUC comparison is more conservative. FiTabR and TabR both obtain a mean value of approximately 0.9510, considerably above the remaining baselines, but the difference between the two retrieval models is negligible. Accordingly, the results do not support a claim that FiTabR improves the ranking capability of TabR on Heart Disease.

A clearer distinction appears in cross-entropy. FiTabR obtains 0.3038±0.0097, compared with 0.3586 ± 0.0267 for TabR and 0.3281 ± 0.0086 for LightGBM. This is the lowest mean cross-entropy in the comparison. Taken together, the Heart Disease results indicate that feature-aware representation improves the classification output and probabilistic quality of the retrieval-based model while preserving the strong ROC-AUC performance of TabR.

5.1.3 Pima Diabetes Results

Performance on Pima Diabetes is less concentrated around a single model. FT-Transformer obtains the highest mean accuracy (0.7701±0.0263), precision (0.7774± 0.0203), recall (0.7701±0.0263), F1-score (0.7722±0.0250), and ROC-AUC (0.8135± 0.0071). LightGBM achieves the lowest cross-entropy at 0.5253 ± 0.0023.

FiTabR is therefore not the strongest model in absolute terms on this dataset. Nevertheless, it improves its direct TabR baseline on four classification metrics. Accuracy increases from 0.7529 to 0.7586, precision from 0.7515 to 0.7595, recall from 0.7529 to 0.7586, and F1-score from 0.7520 to 0.7589. Cross-entropy is also reduced from 0.5511 to 0.5474.

The mean ROC-AUC changes from 0.7963 ± 0.0072 for TabR to 0.7953 ± 0.0047 for FiTabR. Given both the small magnitude of the difference and the reported variability, the results should be interpreted as showing no meaningful improvement in ranking performance. Thus, on Pima Diabetes, FiTabR provides a modest improvement in threshold-based classification behavior but remains behind the strongest alternative baselines.

Table 5.4: Minority-class performance of TabR and FiTabR on the Stroke Prediction dataset.

Model	Pos. Prec.	Pos. Rec.	Pos. F1	Macro F1	Bal. Acc.	ROC-AUC
TabR	0.0000 ± 0.0000	0.0000 ± 0.0000	0.0000 ± 0.0000	0.4873 ± 0.0000	0.5000 ± 0.0000	0.8136 ± 0.0078
FiTabR	0.0000 ± 0.0000	0.0000 ± 0.0000	0.0000 ± 0.0000	0.4873 ± 0.0000	0.5000 ± 0.0000	0.8095 ± 0.0126

5.1.4 Stroke Prediction and Minority-Class Behavior

The Stroke Prediction dataset presents a substantially different evaluation scenario because of its severe class imbalance. Although the aggregate accuracy is approximately 0.95 for TabR and FiTabR, this value alone does not describe the ability of the models to detect Stroke cases.

To examine this issue directly, Table 5.4 reports positive-class and class-balanced metrics for the two retrieval models.

The minority-class results reveal that both TabR and FiTabR fail to predict positive Stroke cases at the default classification threshold. Positive precision, recall, and F1-score are all zero, while balanced accuracy is exactly 0.5000 and macro F1-score is 0.4873. Consequently, the approximately 95% aggregate accuracy of both models is driven by correct prediction of the dominant negative class rather than successful discrimination of both classes.

This distinction is important when interpreting Table 5.3. The identical aggregate accuracy and F1-score of TabR and FiTabR should not be interpreted as evidence of strong clinical classification performance. FiTabR slightly improves cross-entropy, decreasing it from 0.1780 ± 0.0050 to 0.1750 ± 0.0091, but does not resolve the minority-class detection failure. Its mean ROC-AUC is also slightly lower than TabR. Therefore, the Stroke experiment identifies a clear limitation of the current FiTabR formulation: feature-aware representation alone is not sufficient to overcome severe class imbalance.

5.1.5 Cross-Dataset Synthesis

Across the three medical datasets, the effect of the proposed feature-aware representation is heterogeneous rather than uniform. Heart Disease provides the clearest evidence of improvement. FiTabR increases accuracy, precision, recall, and F1-score relative to TabR, substantially lowers cross-entropy, and preserves the same strong ROC-AUC. On Pima Diabetes, the improvement is smaller: FiTabR increases the threshold-dependent classification metrics and slightly reduces cross-entropy, but does not improve ROC-AUC and remains behind FT-Transformer and selected GBDT baselines on several metrics.

Stroke Prediction represents the most difficult case. FiTabR does not improve the aggregate classification metrics over TabR and does not resolve the failure to identify minority-class examples at the default decision threshold. The result demonstrates that improvements in feature representation cannot by themselves compensate for extreme imbalance in the target distribution.

Overall, the experiments support FiTabR as a targeted extension of TabR rather than a universally superior replacement. The feature-aware mechanism provides its strongest benefit on Heart Disease, produces modest gains on Pima Diabetes, and has limited effect on Stroke Prediction. Importantly, the comparison with the direct TabR baseline shows that improvements can be obtained without replacing the underlying retrieval framework. At the same time, the variation across datasets indicates that the effectiveness of the proposed representation depends on the characteristics of the underlying classification problem.

The benchmark results establish the overall predictive behavior of FiTabR, but they do not identify which elements of the feature-aware representation are re-

sponsible for the observed changes. The following ablation study therefore isolates residual preservation, fixed attention scaling, and learned nonuniform weighting. A subsequent feature-attention analysis then examines the model-internal emphasis assigned to the numerical variables, without treating the attention values as causal or clinical importance scores.

5.2 Ablation Study

Recall that FiTabR modifies the numerical representation before the TabR encoder and retrieval stage. As described in the Methodology section, for the i-th sample, the feature-attention module produces the sample-specific attention vector a_i from the numerical input x_i^{num}. The attention vector is scaled as

$$a_i' = \alpha a_i, \qquad \alpha = 2.0, \tag{5.1}$$

and the complete FiTabR model applies residual feature weighting as

$$\tilde{x}_i^{\text{num}} = x_i^{\text{num}} \odot \left(1 + a_i'\right), \tag{5.2}$$

where $\odot$ denotes element-wise multiplication.

To determine which components of this representation mechanism are important, we evaluate three ablated configurations. The first removes the residual term, the second removes the fixed attention scaling coefficient, and the third replaces the learned sample-specific attention vector with uniform feature weights. TabR is included as the reference configuration. The resulting numerical representations are

$$
\begin{aligned}
\text{TabR:} \quad & \tilde{x}_i^{\text{num}} = x_i^{\text{num}}, \\
\text{FiTabR:} \quad & \tilde{x}_i^{\text{num}} = x_i^{\text{num}} \odot (1 + \alpha a_i), \\
\text{No Residual:} \quad & \tilde{x}_{i,\text{NR}}^{\text{num}} = x_i^{\text{num}} \odot (\alpha a_i), \\
\text{No Scale:} \quad & \tilde{x}_{i,\text{NS}}^{\text{num}} = x_i^{\text{num}} \odot (1 + a_i), \\
\text{Uniform:} \quad & \tilde{x}_{i,\text{U}}^{\text{num}} = x_i^{\text{num}} \odot \left(1 + \alpha a_i^{\text{U}}\right),
\end{aligned}
\tag{5.3}
$$

where

$$
a_i^{\text{U}} = \left[\frac{1}{F}, \ldots, \frac{1}{F}\right], \tag{5.4}
$$

and F denotes the number of numerical features. All other components of TabR, including the encoder, candidate retrieval, contextual aggregation, and prediction head, are kept unchanged. Therefore, the experiment isolates the effect of the proposed feature-weighting mechanism.

For each configuration, three independent five-model ensembles are evaluated, covering 15 trained models in total. The reported values represent the mean and sample standard deviation across the three ensembles.

5.2.1 Heart Disease

Table 5.5 reports the ablation results on the Heart Disease dataset. The component columns indicate which elements of the complete FiTabR representation are retained in each configuration.

The Heart Disease results show the clearest effect of the proposed representation mechanism. FiTabR improves the mean accuracy from 0.8841 for TabR to 0.8937 and the F1-score from 0.8844 to 0.8940. The ROC-AUC remains essentially unchanged, while cross-entropy decreases substantially from 0.3586 to 0.3038.

Table 5.5: Ablation study on the Heart Disease dataset. Results are reported as mean ± standard deviation across three five-model ensembles.

Model	Learned a_i	Residual	$\alpha = 2$	Acc. ↑	F1 ↑	AUC ↑	CE ↓
TabR	×	×	×	0.8841 ± 0.0000	0.8844 ± 0.0000	0.9510 ± 0.0006	0.3586 ± 0.0267
FiTabR	✓	✓	✓	0.8937 ± 0.0042	0.8940 ± 0.0042	0.9510 ± 0.0009	0.3038 ± 0.0097
No Residual	✓	×	✓	0.8647 ± 0.0111	0.8651 ± 0.0110	0.9402 ± 0.0061	0.4050 ± 0.0603
No Scale	✓	✓	×	0.8986 ± 0.0072	0.8988 ± 0.0072	0.9508 ± 0.0016	0.3028 ± 0.0070
Uniform	×	✓	✓	0.8913 ± 0.0126	0.8916 ± 0.0125	0.9509 ± 0.0022	0.3240 ± 0.0209

Removing the residual term causes the largest deterioration among the evaluated configurations. Accuracy decreases to 0.8647, F1-score to 0.8651, and ROC-AUC to 0.9402, while cross-entropy increases to 0.4050. This result indicates that retaining the original numerical representation while applying attention-based modulation is important for the Heart Disease dataset.

The No Scale configuration obtains the numerically highest accuracy and F1-score, as well as the lowest mean cross-entropy. However, the difference between No Scale and complete FiTabR is small relative to the variation across ensembles. Thus, the experiment does not provide evidence that the fixed coefficient $\alpha = 2.0$ improves performance. Instead, the learned residual weighting remains effective when the fixed scaling term is removed.

Uniform weighting is also competitive with the complete model. This suggests that part of the gain on Heart Disease may originate from numerical feature modulation itself rather than exclusively from the learned sample-specific attention distribution. Nevertheless, the Uniform configuration exhibits greater variability than the complete FiTabR model.

Figure 5.1 provides a visual comparison of the main Heart Disease ablation results.

Table 5.6: Ablation study on the Pima Diabetes dataset. Results are reported as mean ± standard deviation across three five-model ensembles.

Model	Learned a_i	Residual	$\alpha = 2$	Acc. ↑	F1 ↑	AUC ↑	CE ↓
TabR	×	×	×	0.7529 ± 0.0050	0.7520 ± 0.0058	0.7963 ± 0.0072	0.5511 ± 0.0120
FiTabR	✓	✓	✓	0.7586 ± 0.0086	0.7589 ± 0.0092	0.7953 ± 0.0047	0.5474 ± 0.0044
No Residual	✓	×	✓	0.7500 ± 0.0086	0.7469 ± 0.0082	0.7994 ± 0.0089	0.5254 ± 0.0165
No Scale	✓	✓	×	0.7471 ± 0.0100	0.7467 ± 0.0104	0.7973 ± 0.0075	0.5393 ± 0.0187
Uniform	×	✓	✓	0.7500 ± 0.0149	0.7501 ± 0.0132	0.7944 ± 0.0044	0.5501 ± 0.0086

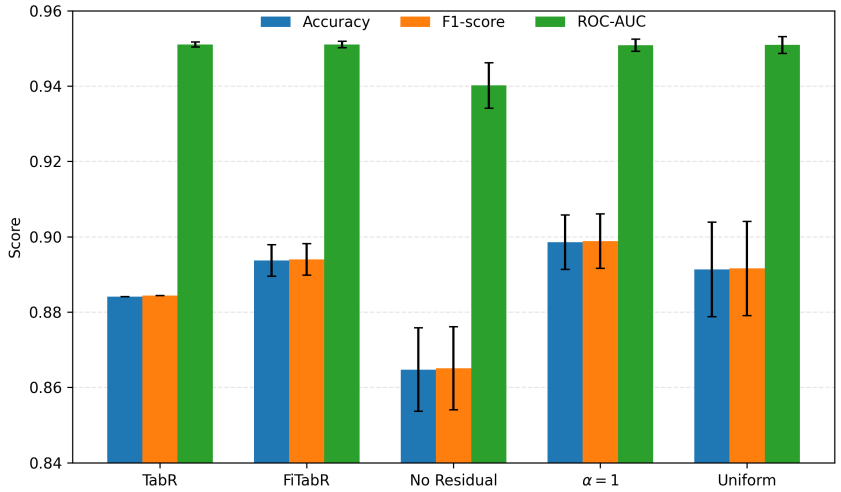


Figure 5.1: Ablation performance on the Heart Disease dataset.

5.2.2 Pima Diabetes

Table 5.6 presents the corresponding ablation results on the Pima Diabetes dataset.

The differences between the Pima Diabetes configurations are substantially smaller than those observed on Heart Disease. Relative to TabR, complete FiTabR increases mean accuracy from 0.7529 to 0.7586 and weighted F1-score from 0.7520 to 0.7589, while cross-entropy decreases from 0.5511 to 0.5474. In contrast, mean ROC-AUC changes slightly from 0.7963 to 0.7953, so the result does not indicate an improvement in ranking performance.

The ablated configurations are similarly close. No Residual obtains the highest mean ROC-AUC (0.7994) and the lowest cross-entropy (0.5254), while No Scale and Uniform remain close to the TabR baseline. In contrast to Heart Disease, the

Table 5.7: Ablation study on the Stroke Prediction dataset. Results are reported as mean ± standard deviation across three five-model ensembles.

Model	Learned a_i	Residual	$\alpha = 2$	Acc. ↑	F1 ↑	AUC ↑	CE ↓
TabR	×	×	×	0.9505 ± 0.0000	0.9263 ± 0.0000	0.8136 ± 0.0078	0.1780 ± 0.0050
FiTabR	✓	✓	✓	0.9505 ± 0.0000	0.9263 ± 0.0000	0.8095 ± 0.0126	0.1750 ± 0.0091
No Residual	✓	×	✓	0.9505 ± 0.0000	0.9263 ± 0.0000	0.8104 ± 0.0089	0.1775 ± 0.0029
No Scale	✓	✓	×	0.9505 ± 0.0000	0.9263 ± 0.0000	0.7985 ± 0.0053	0.1806 ± 0.0004
Uniform	×	✓	✓	0.9505 ± 0.0000	0.9263 ± 0.0000	0.7727 ± 0.0363	0.1801 ± 0.0120

experiment therefore does not identify one component whose benefit is consistent across the reported metrics. The results instead suggest that the effect of feature weighting is weaker and more metric-dependent on this dataset.

Figure 5.2 summarizes the corresponding ablation comparison for Pima Diabetes.

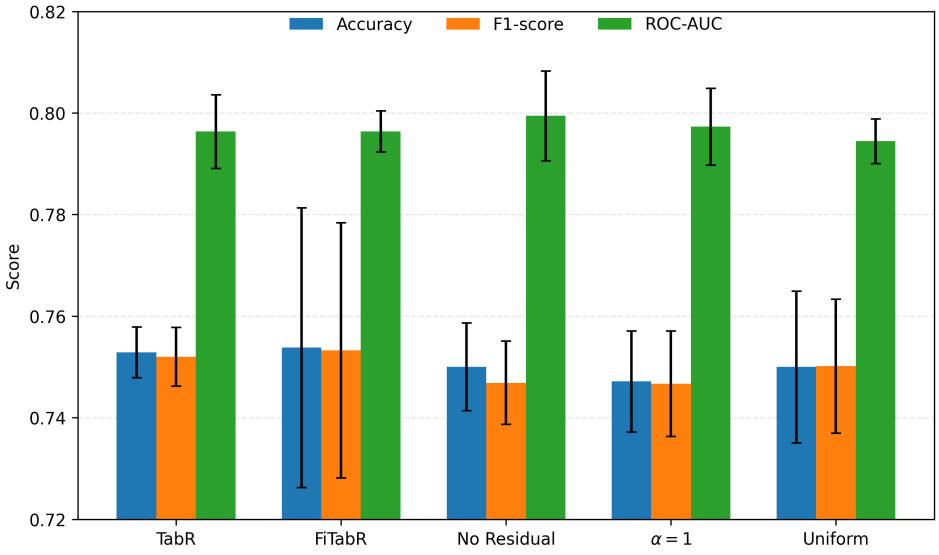


Figure 5.2: Ablation performance on the Pima Diabetes dataset.

5.3 Stroke Prediction

Table 5.7 reports the ablation results on the Stroke Prediction dataset.

The threshold-based metrics are identical across all configurations on the Stroke Prediction dataset. Consequently, accuracy and weighted F1-score do not distinguish between the proposed feature-weighting variants.

Differences are instead observed in probability-based metrics. TabR obtains the highest mean ROC-AUC of 0.8136, while complete FiTabR obtains 0.8095. No Residual remains close to both configurations at 0.8104. In contrast, removing the scaling coefficient reduces ROC-AUC to 0.7985, and replacing learned attention with uniform weighting produces the largest deterioration, reaching 0.7727.

Complete FiTabR obtains the lowest mean cross-entropy, decreasing from 0.1780 for TabR to 0.1750. However, this improvement in probabilistic fit does not correspond to an improvement in ROC-AUC or threshold-based classification performance.

Thus, unlike Heart Disease, the Stroke Prediction experiment does not provide evidence that feature-aware weighting improves overall discrimination. The result also illustrates that weighted threshold-based metrics can conceal differences between models on a strongly imbalanced dataset.

Figure **??** visualizes the ablation results on the Stroke Prediction dataset.

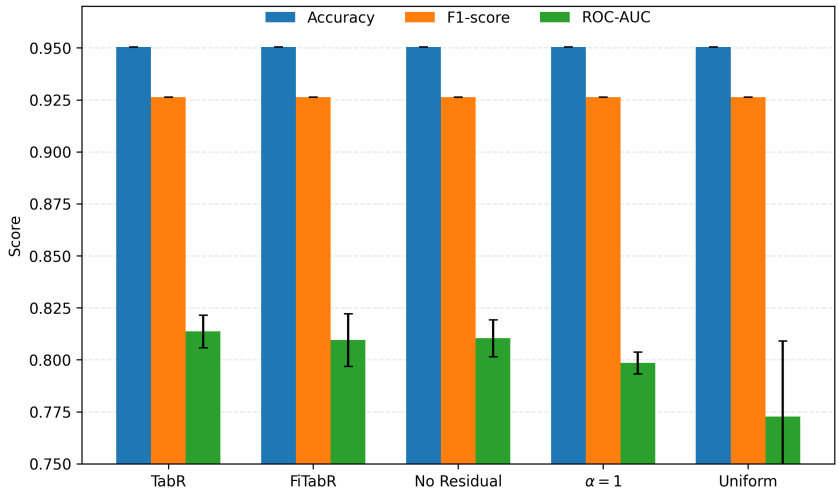


Figure 5.3: Ablation performance on the Stroke Prediction dataset.

5.3.1 Discussion

Overall, the ablation study shows that the effect of the FiTabR components depends strongly on the dataset. The clearest result is observed on Heart Disease. On this dataset, complete FiTabR improves over TabR, whereas removing residual feature weighting causes a substantial performance drop. This provides evidence that preserving the original numerical representation while applying feature-dependent modulation is important for the proposed approach.

The results provide weaker support for the fixed scaling coefficient. The No Scale configuration remains competitive with, and is numerically slightly better than, complete FiTabR on Heart Disease. Since the difference is small relative to the reported ensemble variation, the experiment does not establish that $\alpha = 2.0$ is preferable to using the unscaled attention distribution.

The Uniform variant provides an additional control for the learned attention mechanism. Its competitive performance on Heart Disease indicates that numerical feature modulation itself may account for part of the observed improvement. However, its substantially lower ROC-AUC on Stroke Prediction shows that uniform reweighting is not consistently beneficial across datasets.

Pima Diabetes shows only small differences between the configurations, while Stroke Prediction shows no improvement in threshold-based classification performance. Therefore, the ablation results do not support a claim that the complete FiTabR configuration is universally superior to TabR.

Instead, the experiments indicate that explicit pre-retrieval feature reweighting can be beneficial under some dataset conditions, with the strongest evidence obtained on Heart Disease. Among the examined design choices, residual feature

preservation receives the clearest empirical support, while the contribution of fixed attention scaling and learned sample-specific weighting is more dataset-dependent.

5.4 Clinical Concordance of Learned Feature Attention

Attention-based tabular models such as TabNet use learned feature-selection signals to expose aspects of model behavior [12]. However, attention weights should not automatically be interpreted as faithful explanations of predictions [?]. Accordingly, the analysis in this section is restricted to qualitative clinical concordance. FiTabR attention is treated as model-internal feature emphasis rather than a causal effect, clinical risk coefficient, or direct measure of medical importance.

The mean feature-attention values are compared with established domain knowledge from cardiovascular, diabetes, and stroke guidelines, the Framingham studies, and large prospective meta-analyses [28, 29, 30, 31, 32, ?, 33]. The attention statistics correspond to the selected FiTabR models recorded in the experiment reports.

5.4.1 Heart Disease

The Heart Disease results show partial agreement between the learned FiTabR attention distribution and established cardiovascular knowledge. Cholesterol and age receive the two highest mean attention values and are also well-established cardiovascular risk-related variables. Resting blood pressure receives intermediate attention despite its strong clinical importance, while Oldpeak and maximum heart rate receive lower attention despite their known prognostic value.

The relatively large standard deviation of cholesterol attention (0.1155) indicates that FiTabR does not emphasize this feature equally for every sample. In-

stead, its contribution to the learned representation varies across patient profiles. Figure 5.4 compares the learned mean attention values with the uniform-attention reference.

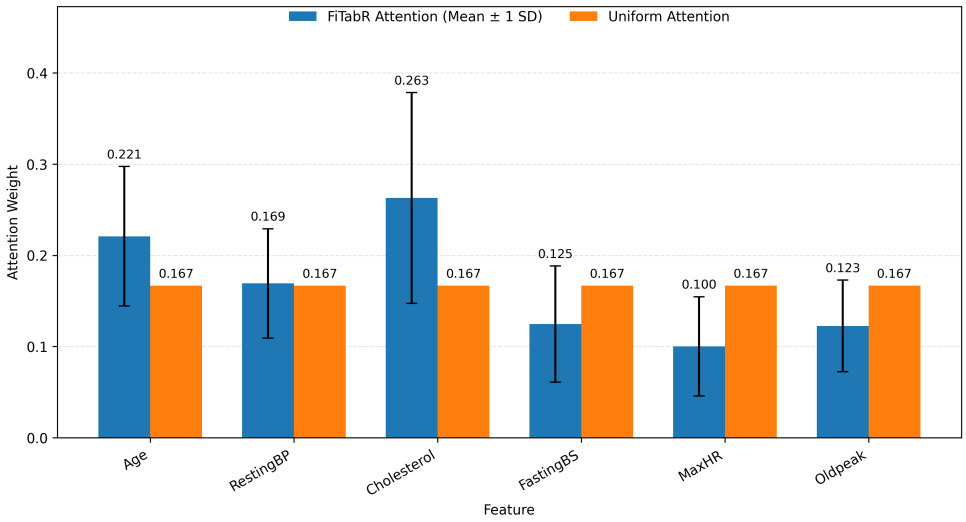


Figure 5.4: FiTabR feature-attention distribution on the Heart Disease dataset. Learned bars show mean attention with ±1 standard deviation; uniform bars show the reference value $1/F$ for each numerical feature.

Table 5.8: Comparison of FiTabR attention with established cardiovascular domain knowledge on the Heart Disease dataset.

Feature	**Attention Mean ± SD**	**Established Clinical Context**
Cholesterol	$\mathbf{0.2627 \pm 0.1155}$	Major contributor to atherosclerotic cardiovascular risk [31, 28].
Age	$\mathbf{0.2208 \pm 0.0765}$	Major cardiovascular risk determinant included in established risk-prediction models [29, 30].
RestingBP	0.1691 ± 0.0600	Elevated blood pressure is a major modifiable cardiovascular risk factor [34, 28].
FastingBS	0.1246 ± 0.0637	Diabetes and elevated glucose are associated with increased cardiovascular risk [32].
Oldpeak	0.1226 ± 0.0502	Exercise-induced ST-segment deviation has established diagnostic and prognostic relevance [35].
MaxHR	0.1002 ± 0.0544	Impaired exercise heart-rate response is associated with adverse cardiovascular outcomes [36].

Overall, the Heart Disease attention pattern is medically plausible in part, but it should not be interpreted as a conventional cardiovascular risk-factor rank-

ing.

5.4.2 Pima Diabetes

The Pima Diabetes results demonstrate partial clinical concordance. SkinThickness receives the highest mean attention, followed by insulin, BMI, and age. These variables are related to adiposity, insulin regulation, and metabolic risk, which are important components of Type 2 diabetes development. The relatively large standard deviation of insulin attention (0.1157) further indicates substantial variation in its model-internal emphasis across individual samples.

The most notable discrepancy is glucose. Although glucose measurements are central to diabetes screening and diagnosis, glucose receives relatively low mean FiTabR attention. BloodPressure and DiabetesPedigreeFunction also receive low attention despite their associations with cardiometabolic and familial risk. Figure 5.5 shows the learned attention distribution against the corresponding uniform reference.

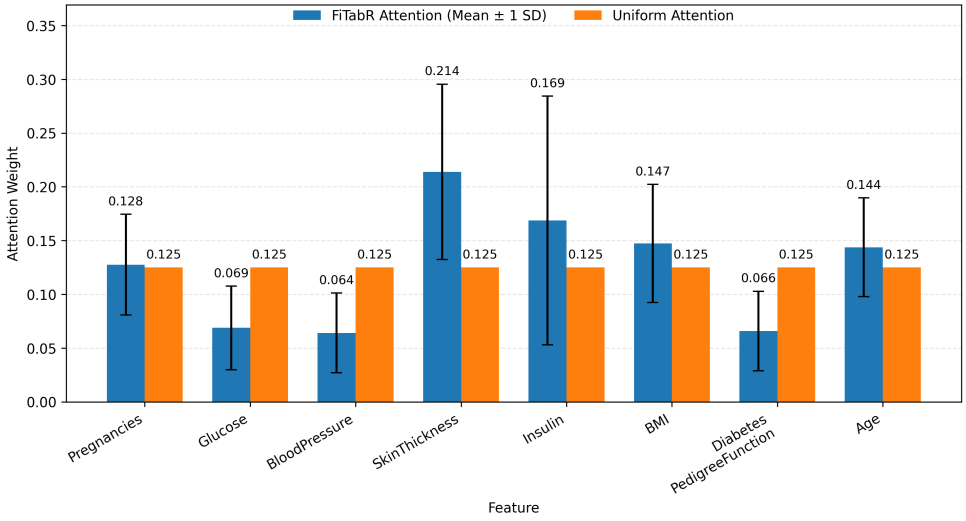


Figure 5.5: FiTabR feature-attention distribution on the Pima Diabetes dataset. Learned bars show mean attention with ±1 standard deviation; uniform bars show the reference value $1/F$ for each numerical feature.

Table 5.9: Comparison of FiTabR attention with established medical domain knowledge on the Pima Diabetes dataset.

Feature	**Attention Mean ± SD**	**Established Clinical Context**
SkinThickness	$\mathbf{0.2139 \pm 0.0816}$	Indicator of subcutaneous adiposity associated with insulin resistance and Type 2 diabetes risk [**?**, **?**].
Insulin	$\mathbf{0.1687 \pm 0.1157}$	Insulin resistance and β-cell dysfunction are central to Type 2 diabetes pathophysiology [**?**].
BMI	0.1474 ± 0.0548	Overweight and obesity are established Type 2 diabetes risk factors [**?**, **?**].
Age	0.1438 ± 0.0459	Increasing age is an established diabetes risk factor used in screening recommendations [**?**].
Pregnancies	0.1276 ± 0.0469	Pregnancy-related metabolic history can be associated with future diabetes risk [**?**].
Glucose	0.0688 ± 0.0389	Glucose measurement is central to diabetes screening and diagnosis [**?**].
DiabetesPedigreeFunction	0.0658 ± 0.0370	Represents familial predisposition; family history is an established diabetes risk factor [**?**].
BloodPressure	$\mathbf{0.0641 \pm 0.0370}$	Hypertension is associated with cardiometabolic and diabetes risk [**?**].

The discrepancy between model attention and established clinical relevance, especially for glucose, reinforces that FiTabR attention is a property of the learned representation rather than a direct ranking of diabetes risk factors.

5.4.3 Stroke Prediction

The Stroke results also demonstrate partial clinical concordance. Average glucose level and age receive the two largest attention values, consistent with established evidence linking diabetes, hyperglycemia, increasing age, and stroke risk. The relatively large standard deviation of glucose attention (0.1262) further indicates substantial variation in its model-internal emphasis across individual samples.

The most notable discrepancy is hypertension. Although hypertension is one of the strongest established modifiable risk factors for stroke, it receives the lowest mean FiTabR attention. The binary hypertension variable contains less informa-

tion than a continuous blood-pressure measurement, and the model may also represent correlated information through age, glucose, BMI, cardiovascular history, and other features. Figure 5.6 visualizes this feature-emphasis pattern.

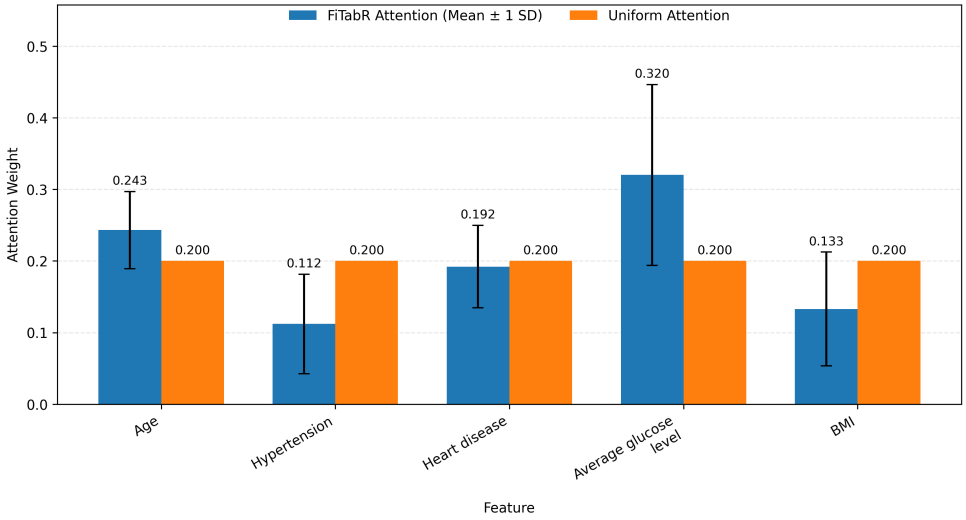


Figure 5.6: FiTabR feature-attention distribution on the Stroke Prediction dataset. Learned bars show mean attention with ±1 standard deviation; uniform bars show the reference value $1/F$ for each numerical feature.

Table 5.10: Comparison of FiTabR attention with established medical domain knowledge on the Stroke Prediction dataset.

Feature	**Attention Mean ± SD**	**Established Clinical Context**
Average glucose level	$\mathbf{0.3200 \pm 0.1262}$	Diabetes and hyperglycemia are established stroke risk factors [32, 33].
Age	$\mathbf{0.2430 \pm 0.0540}$	Major non-modifiable determinant of stroke risk [37].
Heart disease	0.1920 ± 0.0576	Cardiovascular disease is associated with increased stroke risk [38].
BMI	0.1329 ± 0.0794	Higher BMI is associated with stroke risk, partly through metabolic risk factors [39].
Hypertension	$\mathbf{0.1121 \pm 0.0695}$	Major modifiable stroke risk factor and prevention target [33, 34].

5.4.4 Attention Analysis Summary

Across the three datasets, FiTabR learns several attention patterns that are consistent with established medical knowledge: cholesterol and age receive the largest

mean attention on Heart Disease, SkinThickness, insulin, BMI, and age receive comparatively high attention on Pima Diabetes, and average glucose level and age receive the largest attention on Stroke Prediction.

At the same time, clinically meaningful discrepancies remain. Glucose receives relatively low attention on Pima Diabetes despite its central diagnostic role, and hypertension receives the lowest mean attention on Stroke Prediction despite being an established modifiable stroke-risk factor. These findings support interpreting FiTabR attention as model-specific feature emphasis, not as epidemiological, causal, or clinical risk estimates.

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국문초록

Title: Feature-Aware Numerical Representation for Retrieval-Augmented Medical Tabular Classification (FiTabr)

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감사의 글

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